JNTUH-USED 29-11-17 PM Set-1

Code No: 117BX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2017
COMPUTER GRAPHICS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Part- A

(25 Marks)

1.a) What are the hardware devices used for computer graphics? [2]
b) What is aliasing and antialiasing? [3]
c) What is covering (exterior clipping)? [2]
d) Distinguish between uniform scaling and differential scaling? [3]
e) What is Bezier Basis Function? [2]
f) Differentiate between interpolation spline and approximation spline? [3]
g) Classify the Visible Surface Detection Methods. [2]
h) What are the Basic illumination Models. [3]
i) Discuss about motion specifications. [2]
j) What is raster animation? [3]

Part - B

(50 Marks)

2.a) Distinguish Raster and Random scan devices.
b) List the input devices. Explain the functionalities of any one input device. [6+4]

OR

3.a) Illustrate the Bresenham line drawing algorithm, digitize the line with endpoints (20,10) and (30,18) and this has a slope of 0.8.
b) Describe boundary fill algorithm? [6+4]

4.a) Rotate a triangle A(0,0) , B(2,2) , C(4,2) about the origin and about P(2, 2) by and angle of 45^0.
b) Give a brief note on two dimensional viewing transformation pipeline? [5+5]

OR

5.a) Derive window to viewport coordinate transformation.
b) Explain Cohen Sutherland and Cyrus-beck line Clipping algorithms. [4+6]

6. Explain in detail about B-Spline curves and surfaces? Derive parametric Bezier curve equation controlled by four points { (2,5,3), (3,-6,8), (1,-2,3), (-4,2,-2)}. [10]

OR

7.a) Explain the concept of parallel projections in 3D.
b) Explain the 3D clipping algorithm for viewing volume. [5+5]

8.a) Explain area subdivision and A-Buffer method?
b) Write short notes on BSP-tree methods? [5+5]

OR

9. Explain polygon Rendering Methods. [10]

10.a) What are the different tricks used in computer graphics animation?
b) What are the animation functions follow the computer animation system? [5+5]

OR

11.a) Give any four real time animation techniques?
b) Mention the salient features of key frame systems? [4+6]

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